Durable Leverage

Volatility Segmentation for Buy-and-Hold Leverage

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Abstract

Durable leverage remains underdeveloped relative to the scale of long-duration capital in global markets. Moderate leverage (1–2×) can enhance long-run compound growth for assets with positive drift, yet today's leverage stack forces a tradeoff between tactical, timing-dependent leverage and structurally expensive set-and-hold leverage.

Margin and perpetual futures both require ongoing margin management and expose holders to liquidation, making them operationally unsuitable for strategic buy-and-hold capital allocation. Leveraged exchange-traded funds (LETFs) come closest to the desired user experience—automated, passively rebalanced exposure—but finance leverage through external hedging counterparties and therefore embed financing costs that compound against the holder over time.

We call leverage durable when an asset's long-run drift dominates its drag: the sum of financing costs and geometric compounding penalties. Existing automated leverage products often fail this test for two structural reasons: (1) leverage is intermediated through external hedging counterparties whose pricing ultimately reflects short-side capital costs, balance-sheet usage, and liquidity/volatility risk—embedding high financing drag; and (2) industry-standard multiples (>2×) push quadratic volatility drag past the point where drift can compensate.

We introduce volatility segmentation, an on-chain protocol that partitions asset volatility into a protected senior tranche and a leveraged junior tranche. This capital structure compresses the financing premium by sourcing leverage from yield-seeking depositors rather than counterparties who are hedging-constrained. The combination of a structural financing advantage and moderate leverage makes the structure compatible with buy-and-hold capital—aligning leveraged exposure with long-duration directional conviction rather than short-term market timing.

We demonstrate the approach on Bitcoin with a 1.33× leverage ratio. **Over 2015–2025, it outperforms 1× spot; by contrast, 2× underperforms and 3× collapses to near zero**. The architecture is extensible to any sufficiently liquid tokenized asset with a reliable price feed, and is strategically compelling where the underlying exhibits positive long-run drift.

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1. Introduction

Leverage is easy to obtain but hard to keep. A position that doubles your exposure for a day is trivial to construct; one that compounds in your favor over a decade is not. We call leverage durable when, over long horizons, the incremental return from leverage—relative to holding the

underlying unlevered—exceeds its costs: volatility drag and financing drag combined, formalized in Section 2 as the Durability Condition.

One might treat 1× exposure—one dollar of exposure per dollar of capital—as the natural baseline. But there is nothing inherently optimal about it. In fact, half a century of portfolio theory [16, 19, 25] implies that for assets with positive long-run returns, the growth-maximizing allocation is often greater than 1×: roughly 1.5–2× for Bitcoin, Gold, and the S&P 500 at current volatilities. Ayres and Nalebuff [2] argue that young investors should lever equity exposure precisely because time diversification favors it; Gayed and Bilello [14] show that systematic leverage on equities can magnify long-run returns when risk is managed. Yet no existing instrument delivers leveraged exposure you can hold passively for years.

A leveraged position left alone drifts from its target: up moves dilute leverage (reducing upside participation), while down moves concentrate it (increasing risk). Maintaining constant leverage therefore requires periodic rebalancing. Two forces then compound against long-horizon leveraged returns:

- **Volatility drag**—the geometric compounding penalty. Like any volatile asset, a rebalanced leverage strategy is subject to compounding drag; at leverage L, this penalty scales approximately with L^2, making high multiples especially punitive.
- **Financing drag**—the cost of borrowed capital, proportional to the amount leveraged.

Together, these forces compound against the position, and too much leverage is far worse than too little [22].

Existing automated leverage products sit on the wrong side of both forces. LETFs (∼$250B AUM [3]) embed swap financing and implementation costs directly into realized returns; these costs do not appear in the stated expense ratio and, in some cases, can exceed it by an order of magnitude—showing up instead as systematic underperformance versus the target multiple. Perpetual futures also embed high financing drag—they have carried funding rates averaging 10–15% annualized for Bitcoin [6]—and maintaining a target leverage ratio requires ongoing margin management under liquidation constraints. Margin borrowing can sometimes be cheap in calm markets, but it is liquidation-constrained and therefore operationally unsuitable for buy-and-hold allocation; rates can range from roughly 4–20% depending on venue and conditions. Each product works for short-term trading; none is built for leverage held passively for years.

At a high level, durable leverage comes down to two questions: *how volatile is the underlying*, and *how expensive is the leverage*. Market conditions set the underlying's drift (μ), volatility (σ), and the risk-free rate (r_f). Product design sets the leverage target (L) and the financing premium (s). Volatility determines how quickly compounding drag accelerates as leverage increases; the financing premium captures who supplies the leverage and what they require to do so. Leverage is durable when the additional drift earned by levering up exceeds the additional drag it introduces

(equivalently, when $R(L) > R(1)$). Achieving durability therefore requires both disciplined leverage targeting and a capital structure that compresses the financing premium.[1]

2. The Anatomy of Leverage Drag

Long-term leveraged returns are eroded by two distinct forces that are often conflated.

2.1 Volatility Drag

Volatility drag is mathematical: the geometric compounding penalty that arises whenever a position is periodically rebalanced to a target leverage ratio. For a daily-rebalanced position with leverage L on an underlying asset with annualized arithmetic mean return μ and annualized standard deviation σ, the annualized growth rate R is approximated by:[2]

$$R \approx L\mu - \tfrac{1}{2} L^2 \sigma^2 \tag{1}$$

The second term, $\frac{1}{2}L^2\sigma^2$, is the volatility drag. It scales *quadratically* with leverage: doubling L quadruples the drag. This relationship is intrinsic to geometric compounding and cannot be eliminated by design.

For Bitcoin ($\sigma \approx 60\%$), a $1\times$ position already loses 18% annually to volatility drag; at $2\times$ the drag reaches 72%; at $3\times$, 162%—exceeding any plausible drift.[3] Lower-volatility assets such as the S&P 500 ($\sigma \approx 19\%$) can sustain much higher leverage before the penalty becomes prohibitive.

2.2 Financing Drag

Financing drag is institutional: the cost of borrowed capital used to produce leveraged exposure, including any premium required by the leverage provider to warehouse hedging, liquidity, and residual risk. It has two components:

- **Amount borrowed (set by leverage ratio L):** a position with leverage L borrows $(L-1)$ dollars per dollar of equity.
- **Cost per dollar borrowed:** the all-in financing rate r, which varies by instrument and market structure.

[1] Here $R(L)$ denotes the long-run compound growth rate of a periodically rebalanced leverage-L position (net of financing and fees), and $R(1)$ the corresponding $1\times$ growth rate.

[2] Avellaneda and Zhang [1] derive this expression in continuous time; Cheng and Madhavan [7] give the discrete-time analogue.

[3] We use $L = 1.33$ as a running example throughout; it lies near the growth-rate optimum for Bitcoin-class volatility (Section 2.3), and Section 3 derives it from the protocol's capital structure.

Total annual financing drag is:

$$\text{Financing drag} \;=\; (L-1)\times r \tag{2}$$

where:

- $r = r_f + s$
- r_f is the risk-free rate
- s is the financing premium above risk-free.

We treat s as a *reduced-form, all-in premium* that subsumes the compensation required for hedging frictions, balance-sheet usage, liquidity and market-impact costs, and any residual risk that cannot be perfectly hedged. In practice, s is determined by market structure and can vary over time, but its average level is the key design variable.

Unlike volatility drag, financing drag is reducible. Lower leverage shrinks the $(L-1)$ term, and redesigning the capital source can compress s. Because volatility drag is intrinsic and cannot be designed away, s is the primary designable cost parameter at a given leverage ratio—making it the main target for protocol optimization.

2.3 The Durability Condition

Combining Equations (1) and (2), the compound growth rate of a leveraged position with financing cost r is:

$$R(L) \;\approx\; L\,\mu \;-\; \tfrac{1}{2}\,L^2\,\sigma^2 \;-\; (L-1)\,r, \tag{3}$$

where $r = r_f + s$.

The first term—drift—grows linearly with leverage. The second and third terms—volatility drag and financing drag—grow quadratically and linearly, respectively. At moderate leverage, drift can dominate. But as L increases, quadratic drag accelerates and eventually overtakes drift. Beyond that crossing point, additional leverage destroys value.

At representative Bitcoin parameters ($\mu = 50\%$, $\sigma = 60\%$), leverage above roughly 1.5–1.6× already underperforms holding the underlying unlevered, and the threshold drops as financing costs rise.

Excess compound growth. Define the *excess compound growth* of leverage L relative to a 1× position as:

$$\Delta R(L) \;:=\; R(L) - R(1). \tag{4}$$

Leverage is *durable* when it adds value versus the 1× baseline, i.e. when $\Delta R(L) > 0$.

The growth-maximizing leverage is the continuous-time Kelly fraction [16, 19, 28]:

$$L^* \;=\; \frac{\mu - r}{\sigma^2}. \tag{5}$$

Durability Condition. A leveraged position is durable if and only if it adds positive excess compound growth:

$$\Delta R(L) \;=\; R(L) - R(1) \;>\; 0.$$

Equivalently,

$$L\,\mu \;-\; \tfrac{1}{2}\,L^2\,\sigma^2 \;-\; (L-1)(r_f + s) \;\;>\;\; \mu \;-\; \tfrac{1}{2}\,\sigma^2.$$

Intuitively, the incremental drift gained from leverage must exceed the incremental drag it introduces.

The condition takes drift μ as an input. When drift is high, the productive band extends well above 1×; when drift is low or negative, L^* falls below 1×—the framework correctly says to de-lever.

Solving the Durability Condition across leverage ratios defines a *productive band*—the range where $\Delta R(L) > 0$. Figure 1 evaluates this relationship across representative (μ, σ) regimes.

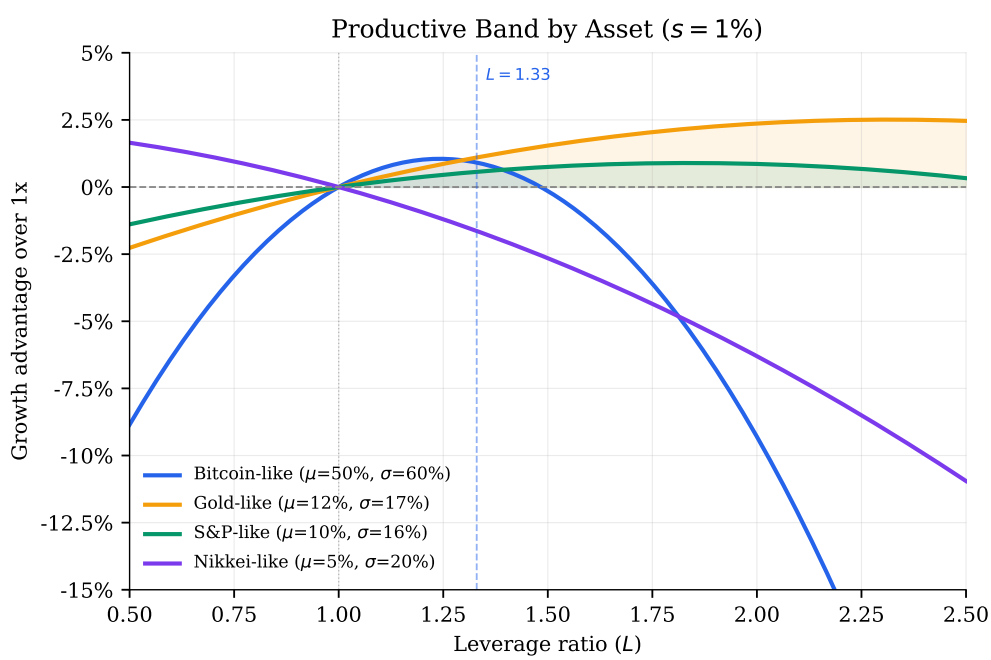

Figure 1: Excess compound growth rate $R(L) - R(1)$ at $s = 1\%$ for four (μ, σ) regimes representative of different asset classes (see text). The shaded region above zero is the *productive band*—leverage adds value only inside it. The dashed line shows $L = 1.33$.

These are closed-form evaluations of Equation (3); Section 4 tests them against actual returns. For Bitcoin-class parameters, $L = 1.33$ sits near the growth-rate optimum and comfortably inside the productive band at $s = 1\%$.

3. Volatility Segmentation

We introduce an on-chain protocol that provides durable, buy-and-hold leverage through volatility segmentation. The structure partitions asset volatility into two perpetual derivatives:

- **Perpetual Senior (Sr)**—a yield-bearing position with volatility protection, provided by Junior first-loss capital.
- **Perpetual Junior (Jr)**—capital-efficient leveraged exposure absorbing first-loss volatility.

Junior capital absorbs underlying price movements first, giving holders leveraged exposure. Senior exposure remains protected until Junior capital is completely depleted. Because leverage arises from the capital structure itself, no external counterparty is needed to take the opposite side of a trade.

Applied to Bitcoin, the structure produces two perpetual derivatives: **Bitcoin Senior (BTC-Sr)**—downside-protected BTC exposure with yield—and **Bitcoin Junior (BTC-Jr)**—moderately leveraged BTC exposure. The remainder of this section walks through how these are configured; Section 4 tests them empirically.

3.1 Capital Structure

The protocol creates time-bounded bonds backed by an underlying asset (e.g. BTC). Each bond is split into a Senior tranche and a Junior tranche according to a configurable *tranche ratio* $\alpha \in (0, 1)$. Senior holds a priority claim; Junior absorbs residual price movement. A concrete example makes the mechanics clear.

Example. Consider a single bond created when BTC trades at \$100,000, with $\alpha = 0.50$. The bond holds 1 BTC of underlying, so the Senior tranche has a \$50,000 claim and the Junior tranche has a \$50,000 claim.

- **BTC rises 10% to \$110,000.** The Senior claim stays at \$50,000 (capped at par). The Junior receives the remainder: \$110,000 − \$50,000 = \$60,000, a 20% gain on its initial \$50,000—exactly 2× the underlying move.
- **BTC falls 10% to \$90,000.** The Senior claim stays at \$50,000 (protected). The Junior receives \$90,000 − \$50,000 = \$40,000, a 20% loss—again 2× the underlying move.

Junior absorbs all price movement on half the capital, producing $\approx 2\times$ leverage without borrowing.

General form. For tranche ratio α, Senior holds a par claim of $\alpha \cdot P_{\text{open}}$ per unit of underlying, where P_{open} is the underlying price at bond inception. At maturity, Senior is paid first up to par; Junior receives the remainder. This gives Junior an effective leverage of $1/(1-\alpha)$ at inception.

Bonds have a finite maturity. Perpetual exposure is achieved by automatically rolling maturing bonds into fresh ones; holders of either tranche maintain uninterrupted exposure without manual intervention.

The target leverage ratio L is configurable per underlying asset by adjusting the *target system ratio*—the ratio of Junior capital to Senior capital in the reserve. The appropriate L is determined by the asset's productive band (Section 2.3): the (μ, σ) regime pins down L^*, and the protocol targets leverage near that optimum. For BTC with $\alpha = 0.50$, the leverage ceiling is $2\times$; the protocol targets $L = 1.33\times$ (system ratio 3:1 Junior-to-Senior), which falls near L^* and well inside the productive band for BTC-class volatility.

3.2 Leverage Target Selection

The cost of getting leverage wrong is radically asymmetric. Undershooting the optimal ratio—holding $1.33\times$ when L^* is $1.5\times$—sacrifices roughly a percentage point of annual return. Overshooting it—holding $2\times$ or $3\times$ when the productive band closes at $1.8\times$—is catastrophic: Section 4.1 shows that $3\times$ BTC is effectively annihilated in the first major drawdown and never recovers. The penalty curve is not symmetric around the optimum; it is gentle on the left and vertical on the right.

This asymmetry explains why every major crypto leverage product destroys long-run value. The industry defaults to 2–$3\times$ because those ratios work in equities, where $\sigma \approx 19\%$ and the productive band extends past $3\times$. But Bitcoin's $\sigma \approx 60\%$ collapses the band to roughly 1.0–$1.6\times$—and $2\times$ is already outside it. The problem is not leverage itself but the uncritical transplant of equity-market intuitions onto crypto-volatility assets.

The protocol targets $1.33\times$ for Bitcoin—a ratio that remains inside the productive band across every non-stress (μ, σ) regime tested in Section 4, from current Bitcoin down to equity-like parameters. The priority is regime-robustness, not point-optimality.

3.3 Funding

Like a perpetual future, the protocol uses a funding transfer to anchor the system to its target. Perp funding settles every eight hours based on the premium to spot; here, Junior pays Senior once daily based on the deviation from the target system ratio. The rate is market-determined:

it emerges from the supply–demand balance between Senior and Junior capital, just as perp funding emerges from the balance between long and short interest.

We estimate the equilibrium BTC-Sr yield at roughly $\approx 4.5\%$ ($r_f \approx 3.5\%$ plus $\sim 1\%$). Section 5.1 derives this estimate from structural comparisons. At a 3:1 system ratio, Seniors earn 4.5% on $100 of capital—$4.50 per year. Spread across $300 of Junior equity, this amounts to an effective financing cost of $\approx 1.5\%$.

System fees are governed by asymmetric, convex curves tied to the system's deviation from its target ratio. Minting or redeeming when the system is balanced costs nothing or near-nothing; transacting in a way that increases imbalance incurs escalating fees. Collected fees go to Junior holders, not to external parties, partially offsetting their financing cost below the $\sim 2.0\%$ headline estimate. For the BTC deployment, the protocol charges an AUM fee of 0.50%. The savings from cheaper financing are so large that Jr could charge several times this fee and still undercut most LETFs on total cost.

4. Empirical Analysis

We backtest the framework against daily prices from January 2015 to December 2025 for Bitcoin modeled using daily rebalancing. Three headline results:

1. **Moderate leverage is optimal for Bitcoin.** Beyond a narrow band, higher leverage destroys wealth (Section 4.1).
2. **The strategy survives bear markets.** Durability holds across full market cycles—bull surpluses more than compensate for bear drawdowns (Section 4.2).
3. **Lower financing drag compounds to materially higher wealth.** BTC-Jr's cost advantage over LETFs and perpetual futures widens with every year of compounding.

The result is robust to assumptions—BTC-Jr outperforms spot with a 3× margin of safety on financing costs—and generalizes across assets and drift–volatility regimes.

4.1 Leverage and Long-Run Wealth

Figure 2 plots the growth of $1 invested in BTC at 1×, 1.33×, 2×, and 3×, all using BTC-Jr parameters (0.50% AUM fee, 4.5% Sr yield)—the cheapest instrument available.

The 1.33× path dominates not just in terminal value but throughout the sample: it overtakes spot early and never relinquishes the lead. The 2× position suffers catastrophic drawdowns during bear markets—the quadratic volatility penalty amplifies losses to the point where the position never fully recovers. At 3×, the penalty is so severe that the position is effectively

annihilated within the first major drawdown. This is the Durability Condition playing out in real time: drift exceeds drag only at moderate leverage.

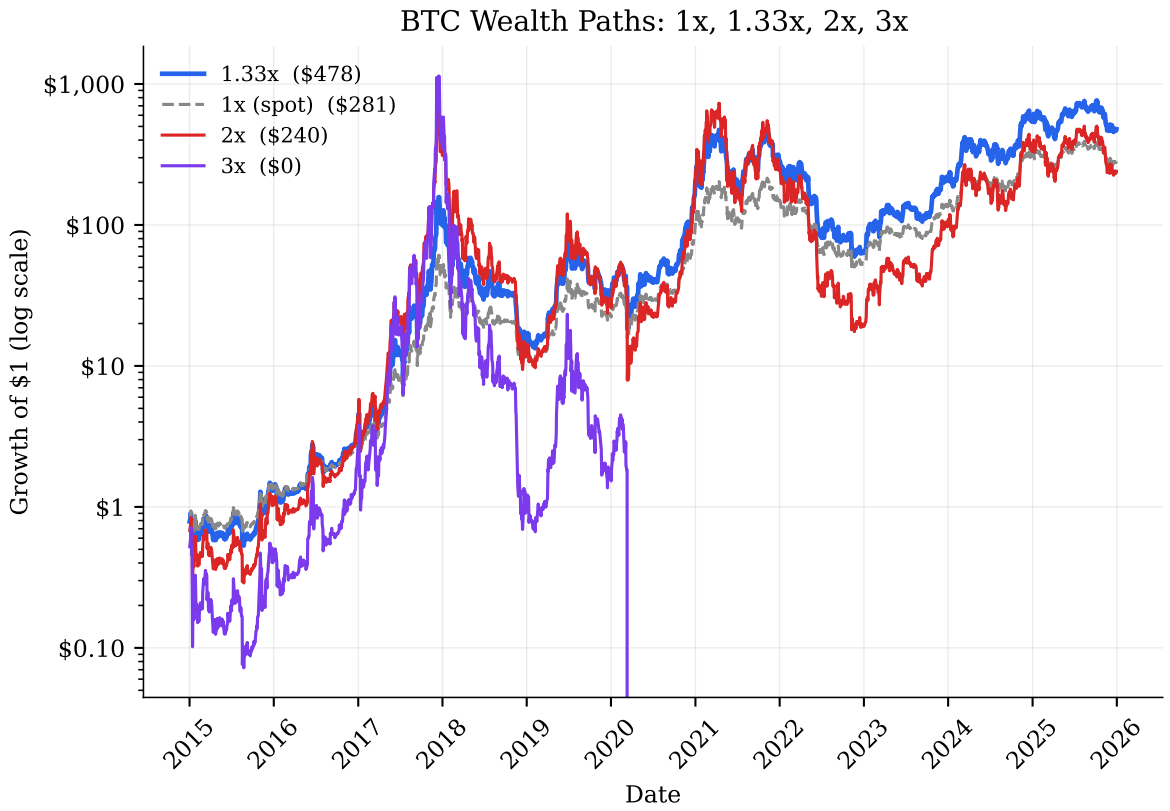

Figure 2: Growth of $1 in BTC at four leverage ratios (2015–2025), all using BTC-Jr parameters. 1.33× dominates throughout; 2× suffers catastrophic drawdowns from which it never fully recovers; 3× is effectively annihilated. Terminal values shown in legend.

4.2 Regime Robustness

A long-duration leverage strategy should not be judged by whether it outperforms in every subperiod. In sustained drawdowns, leveraged exposure will typically underperform spot because it magnifies losses and drag accumulates precisely when capital is shrinking. The relevant question for strategic holders is therefore whether the *compounding surplus* earned in up and trending environments outweighs the *compounding deficit* incurred in down environments over a full market cycle.[4]

Put differently: the strategy is durable if it can compound faster than spot across complete cycles—even if it predictably lags during drawdowns.

[4]Regimes are classified by manual date ranges aligned to major market turning points (e.g. the November 2021 all-time high marks the start of the 2022 bear market). See Figure 3 for the exact boundaries.

Over the full 2015–2025 period, the surplus from up environments dominates the deficit from down environments, so the strategy outcompounds spot in aggregate.

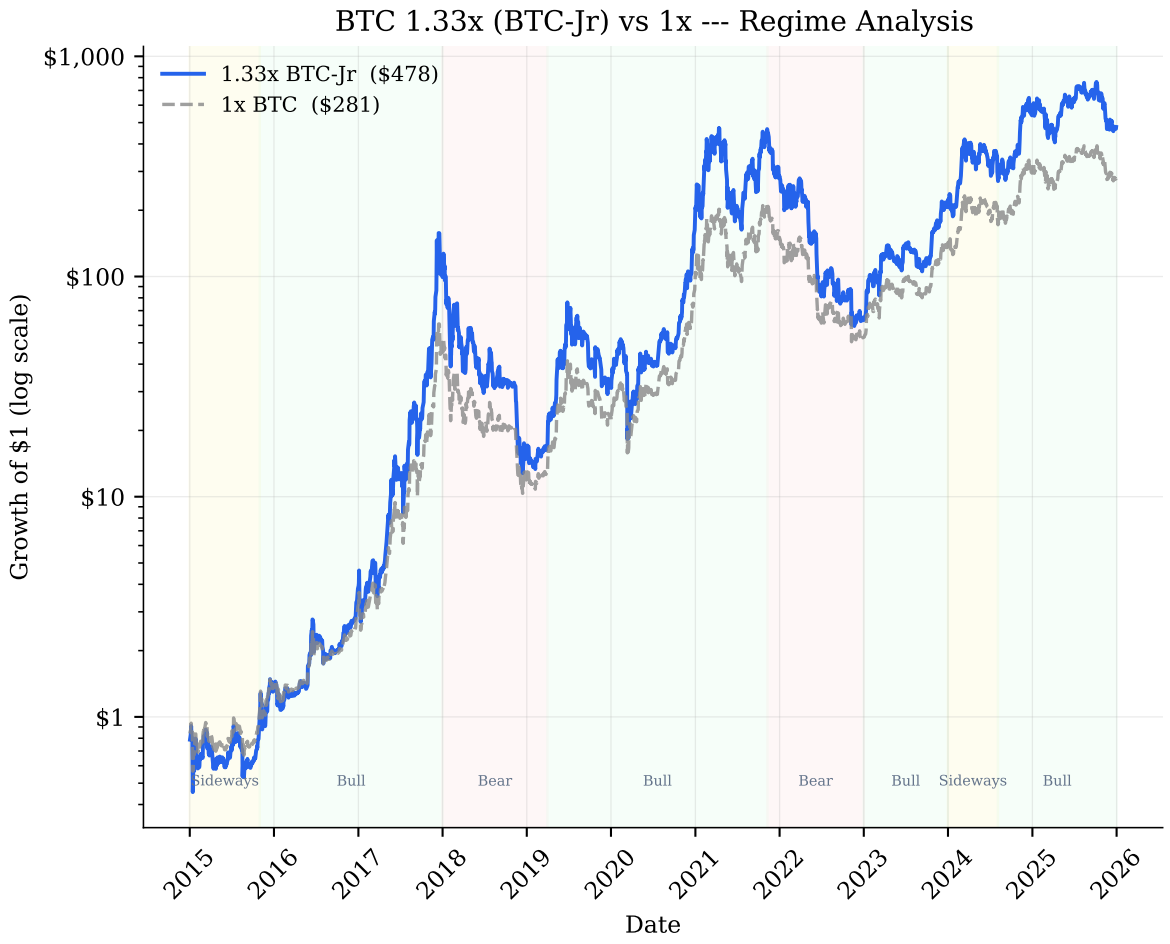

Figure 3: BTC 1× vs 1.33× (BTC-Jr) with regime shading. Green: bull regimes. Yellow: sideways. Red: bear regimes.

To examine these dynamics more concretely, we zoom into three representative periods: a bull-dominated stretch (January 2023 to December 2025), a sustained down market (November 2021 to December 2022), and a sideways/range-bound market (January to August 2024).

Bull-dominated. Over Jan 2023–Dec 2025, $10,000 in BTC grows to $53,200; the same capital in BTC-Jr reaches $75,100—41% outperformance despite the additional drag.

Down market. During the 2022 crypto winter, BTC-Jr draws down to $1,370 vs. BTC's $2,480—an extra ∼11 percentage points of additional drawdown, the premium paid for bull-market outperformance. Crucially the drawdown is *painful but not terminal*: modest leverage preserves enough capital for the subsequent recovery.

Sideways market. In the range-bound first half of 2024, BTC-Jr holds at $13,700, broadly tracking the underlying. By contrast, a 3× LETF decays to $10,000 —and real-world products like the GraniteShares 3× Long MicroStrategy ETP [13] suffer even more severe drag—becoming effectively unholdable.

Taken together, the regime decomposition and the per-condition close-ups confirm the main empirical claim: 1.33× BTC-Jr is *durable*—it compounds in trending markets, survives drawdowns, and avoids decay in sideways conditions.

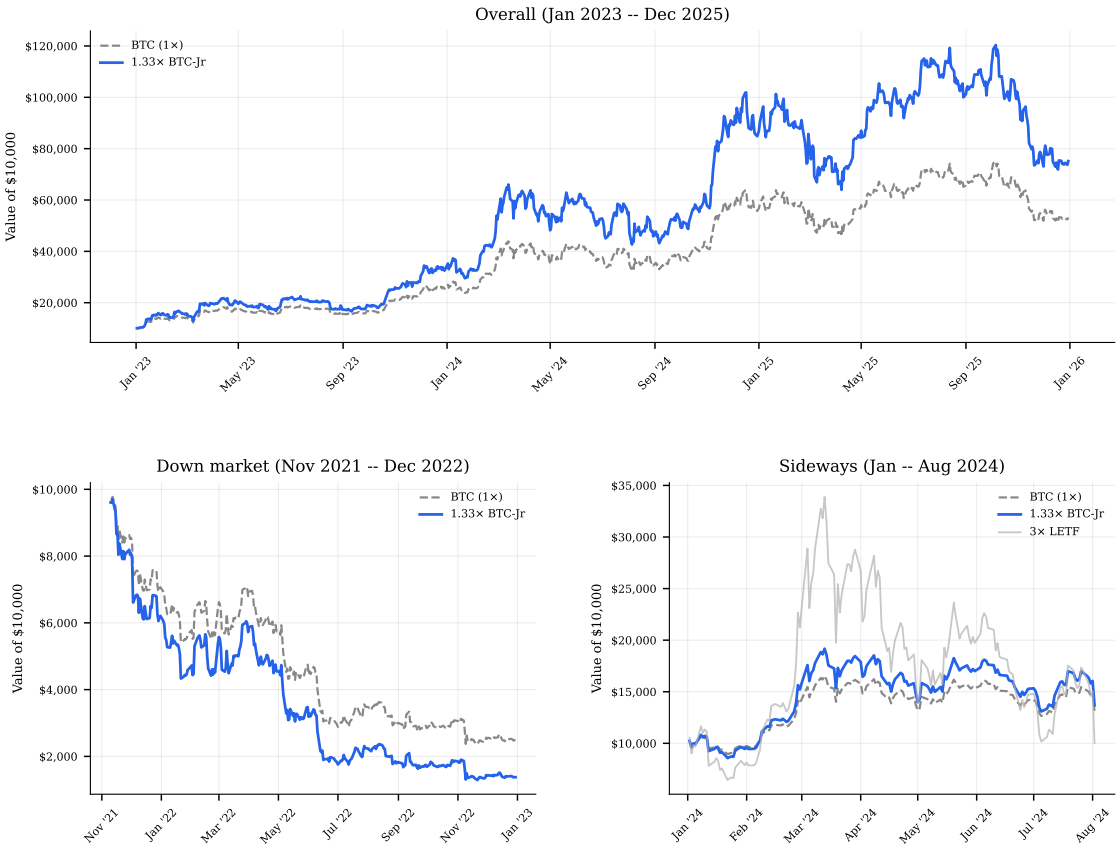

Figure 4: Value of $10,000 invested in BTC (1×) vs 1.33× BTC-Jr across three market conditions. **Top:** Overall (Jan 2023 – Dec 2025). **Bottom left:** Down market (Nov 2021 – Dec 2022). **Bottom right:** Sideways (Jan – Aug 2024), with a 3× LETF shown for comparison. All simulations use 4.5% Sr yield and 0.50% AUM fee.

5. Economics

The preceding analysis establishes the empirical case. This section examines the structural economics: why the capital is cheaper, why moderate leverage is optimal, where the architecture applies, and where it does not.

5.1 Capital Structure Advantage

The primary innovation is not the leverage ratio but the *source* of leverage. Every existing instrument couples its financing premium to the hedging costs of a specialized counterparty—and for Bitcoin, those costs are extreme. Volatility segmentation replaces these counterparties with yield-seeking depositors, lowering the financing premium by an order of magnitude and compounding into a material wealth gap over time.

Why is this capital cheaper? The leverage providers behind automated products must actively hedge and consume balance sheet; those costs flow through as financing premia. A Senior depositor, by contrast, benchmarks yield against treasuries—not swap-dealing returns—so the required credit spread is small. The supply pool is also wider: yield-seeking capital is a broader base than regulated dealer balance sheets, and on-chain settlement extends it globally.

The Sr yield used throughout this paper ($r_f + \sim 1\%$, or $\approx 4.5\%$ at current rates) is our best estimate, not an observed market rate; supply and demand will set the actual number. Early yields will likely be higher—the market will price novelty risk—then compress as capital scales in. (Ethena's sUSDe, a tokenized yield product, converged toward the risk-free rate after supply grew from \$85M to \$12B [8, 9].)

Structural comparisons help bound the range. With $\alpha = 0.5$, Seniors sit above 50% Junior subordination on short-dated collateral, which resembles a conservative secured-credit profile. In traditional markets, adjacent risk tiers price close to the risk-free rate; on-chain, large lending venues often clear near r_f for overcollateralized lending with *no* subordination. Seniors here receive strictly stronger protection. Taken together, these benchmarks suggest a mature-protocol premium on the order of $\sim 1\%$, with early deployments potentially wider until novelty risk compresses.

Existing leverage products fail the Durability Condition for the reasons identified in Sections 3.2 and 2.2: leverage ratios imported from equity markets fall outside the productive band for Bitcoin-class volatility; LETFs embed hidden swap costs that dwarf their stated expense ratios; and perpetual futures charge funding on the full notional—not just the borrowed portion. In each case, total drag exceeds the drift from leverage.

5.2 Moderate by Design

The $1.33\times$ target is not an optimization residual—it is determined by the capital structure: a 3:1 system ratio at $\alpha = 0.50$ yields $L = 1.33$ (Section 3), independent of any backtest. At Bitcoin-class volatility, the penalty for overshooting the productive band is orders of magnitude larger than the penalty for undershooting it (Section 3.2), which means the regime-robust choice sits at or below the point estimate of L^* across the range of plausible regimes. For high-volatility assets, moderate leverage is not a compromise but the structural optimum—the ratio that compounds

reliably across the widest range of market conditions.

The structure is also self-dampening under stress. If Senior depositors withdraw during a drawdown, the system ratio shifts toward Junior capital, which mechanically *reduces* leverage—Junior exposure moves closer to $1\times$ spot. The risk is asymmetric in the safe direction: stress compresses leverage rather than amplifying it. Junior holders may temporarily receive less leverage than they pay for, but the capital structure prevents the dangerous scenario—leverage spiking precisely when drawdowns are deepest.

5.3 Beyond Bitcoin

The mechanism generalizes beyond Bitcoin. Volatility segmentation does not rely on Bitcoin-specific market structure; it relies on a generic property of leveraged compounding: the value of leverage depends on the underlying drift–volatility regime and on the all-in financing premium. Where the underlying exhibits positive long-run drift, and where the financing premium can be compressed, moderate leverage can sit inside the productive band and remain compatible with long-horizon allocation.

The more interesting question is why this regime has been so difficult to serve. Historically, "durable leverage" was largely out of reach because the standard way to industrialize leverage is through deep derivatives infrastructure. In traditional markets, efficient, continuously rebalanced leverage depends on the ability of intermediaries to warehouse and hedge risk at scale—which, in turn, requires extremely deep futures, options, and swap markets. Those markets take years to develop and often never develop at meaningful depth; when they do, it is typically for major indices and a small set of benchmark underlyings. Outside those venues, the financing premium remains structurally high and unstable, and continuously rebalanced leverage tends to sit outside the productive band.

Given that constraint, the market optimized for a different customer and a different objective: tactical short-horizon leverage. At trading horizons, users are not especially sensitive to compounding efficiency; directional P&L dominates, and the product can tolerate wide premia and implementation friction. The result is an ecosystem of leverage tools designed for speculation rather than long-duration capital allocation.

Volatility segmentation changes the feasibility frontier. Because the leverage is embedded in a capital structure that can be implemented on-chain, it does not require pre-existing, institution-scale derivatives markets to approach the productive band. A reliable price feed and a depositor base are sufficient. This opens a new product category: moderate, automated leverage engineered for buy-and-hold investors, not just short-term traders.

6. Limitations

1. **The financing premium is held constant.** The simulations use historical risk-free rates but hold the financing premium s constant; in practice, s varies with market conditions. This constant-premium approach is standard in academic LETF models [1, 7] but does not capture procyclical financing spikes.

2. **Outperformance depends on realized drift.** Durability is defined relative to holding the underlying at 1×: leverage adds value only when incremental drift over the holding period exceeds the incremental drag from compounding and financing. The protocol can be designed to minimize avoidable drag and to expand the range of positive-drift environments in which modest leverage outperforms. But if the underlying's realized drift over the holding period is persistently negative, long leverage will underperform the 1× baseline by construction. The framework handles any μ—including negative drift—but this paper scopes to buy-and-hold assets with positive secular drift.

3. **Smart-contract and oracle risk add a yield premium.** Any on-chain protocol carries smart-contract and oracle risk that depositors price into required yields. DeFi insurance costs (a market-implied proxy) range from 1–3% annually for mature, battle-tested protocols such as Aave and Compound [21], and realized loss rates across DeFi lending have fallen from ∼30% in 2020 to ∼0.5% in 2024—a 98% decline [17] as audit practices, formal verification, and operational track record accumulate. For a new protocol, this premium is real and may initially widen the effective Sr yield above the 4.5% estimate. As the protocol matures and builds Lindy time (credibility that accumulates with operational track record), the premium should compress toward zero.

7. Conclusion

Leverage is durable when incremental drift exceeds drag. Every leveraged product pays a cost to stay leveraged. You cannot avoid volatility drag—that is physics—but you can choose who you borrow from and how much you pay. Today's options are expensive. LETFs hide double-digit swap rates inside an expense ratio. Perpetual futures charge funding on the full notional, not just what you borrowed. Margin loans price in the lender's worst case. At Bitcoin volatilities, these costs eat most or all of the extra return leverage is supposed to deliver.

Volatility segmentation changes the economics. Depositors, not dealers, supply the capital, and they accept a fixed yield instead of demanding a risk premium. The financing cost drops by an order of magnitude. That difference—a few percent a year—compounds quietly in the background, and over a decade it separates a position that builds wealth from one that merely matches spot.

The mechanism is simple: a price feed and a depositor base. No derivatives market required. Any asset with positive long-run drift clears the bar, which means the same structure works beyond crypto. The point of this paper is that durable leverage is now a design choice, not a market constraint.

The companion Supplementary Material [27] provides multi-asset simulations, LETF cost decomposition, fee capacity analysis, and a leverage instrument survey.
https://drive.google.com/file/d/1VPu2SOqdjIAz22j7rs_k-DOupghCBoWm/view

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